

Probabilistic County-Scale Prediction of Storm-Driven Power Outages and the Decision Value of Ensemble Forecasts over Michigan

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Abstract. Distribution-system power outages are the weather impact that electricity customers actually experience, yet operational planning questions — how much outage consequence should be expected in each county, and whether a forecast is good enough to act on — are seldom answered probabilistically. We present a reproducible, end-to-end framework that links ERA5 reanalysis and NOAA GEFS ensemble forecasts to county-day outage records from the EAGLE-I archive across Michigan for 2018–2023. A three-part hurdle model estimates (i) the probability that a county-day experiences an outage event, (ii) the conditional distribution of customer-hours lost, and (iii) the conditional restoration duration under right-censoring; Monte Carlo composition combines the three into a predictive distribution of consequence that preserves zero inflation and heavy tails. The model was frozen on pre-2023 data and scored *once* on the full 2023 holdout. On that holdout the occurrence model attains a Brier score of 0.0795, improving on county climatology (0.0859), a reduced logistic GLM (0.0840), and a 20 m s^{-1} gust-threshold rule (0.1118); magnitude skill relative to climatology remains positive (CRPS skill +0.084) and restoration concordance is 0.619. Two 2023 GEFS case studies isolate where the framework is and is not reliable: all four lead times of an August convective wind event contained the observation within their 10–90% predictive intervals, whereas every lead time of a February ice storm fell below it. A cost–loss analysis translates calibrated probabilities into decision value; at a cost–loss ratio of 0.10 and a five-day lead, triggered counties contained 99% of the realized event loss. The framework is region-portable through a single configuration file, and all interruption-cost figures are explicitly flagged as placeholders pending utility-specific inputs.

Keywords: power outage prediction; probabilistic forecasting; ensemble forecast value; ERA5; GEFS; EAGLE-I; cost–loss analysis; energy resilience.

1 Introduction

Severe weather is the dominant cause of sustained interruptions on electricity distribution systems, and the consequence borne by customers is measured not in damaged assets but in service lost. Utilities and emergency managers therefore face a forecasting problem that is only partly meteorological: given the atmospheric state expected over a region, how much outage consequence should be anticipated, where, and is the anticipated consequence large enough and certain enough to justify pre-positioning crews, requesting mutual aid, or issuing public advisories?

A substantial literature addresses the first half of this question. Statistical outage-prediction models have been developed for tropical cyclones (Guikema et al., 2014; Nateghi et al., 2014) and for extratropical storms in the northeastern United States (Wanik et al., 2015; Cerrai et al., 2019), generally by regressing outage counts on hazard, land-cover, and network covariates. These studies established that weather-conditioned statistical models can substantially outperform climatological and threshold baselines. Two features of the operational problem have received less attention. First, most

published models target a conditional *mean* or a deterministic count, whereas the quantity a planner needs is a full predictive *distribution*: outage consequence is zero-inflated and heavy-tailed, and the decision-relevant information sits in the upper tail. Second, statistical skill is usually reported without translating it into decision value, so it remains unclear whether a given level of skill would have changed any action.

This paper addresses both. We construct a county-day probabilistic model of outage consequence for Michigan over 2018–2023 and evaluate it under a strictly frozen experimental design, then carry the calibrated probabilities through a cost–loss decision framework (Richardson, 2000; Zhu et al., 2002) to quantify the operational value of acting on them.

Three design commitments shape what follows.

Consequence, not fragility. The target is what customers experienced — probability of an outage event, customer-hours lost, and restoration duration — and not the physical failure of any asset. County-level outage records contain no information about which pole or conductor

failed, and we make no claim about physical damage.

Distributions, not point forecasts. Every stage emits a predictive distribution, and every headline metric is a proper scoring rule (Gneiting and Raftery, 2007) or a calibration diagnostic (Gneiting et al., 2007), evaluated against explicitly stated reference forecasts.

One test, scored once. The temporal split was fixed before the data were examined. The 2023 holdout was opened a single time, after the model, the calibrator, the reporting cohort, and the configuration hash were frozen. No 2023 outcome informed tuning, calibration, bias correction, or model selection.

Section 2 describes the inputs; Section 3 the model and verification framework; Section 4 the frozen-model results; Section 5 the decision-value analysis; and Section 6 what the results do and do not support.

2 Data

2.1 Outage observations

Outage consequence is derived from the EAGLE-I archive of recorded county-level electricity outages (Brelsford et al., 2024), which reports the number of customers without service by county at approximately 15-minute resolution. Records are resampled to hourly county series and normalized by Monthly County Customer (MCC) counts to give an outage fraction f_{ct} for county c at hour t .

Two properties of this archive drive the preprocessing. Reporting coverage is not uniform across counties or years, so the study defines a *frozen reporting cohort*: counties with stable annual coverage across the training and validation years. Counties whose coverage changes within that window are excluded, and any coverage gap confined to the test year is recorded as a stated limitation rather than used to alter the cohort — changing the study population between fitting and testing would make the holdout something other than a held-out test. Second, a nonzero baseline of routine, non-storm outages is always present. We remove it with a rolling 30-day 10th-percentile baseline per county and work with the excess fraction $f_{ct}^{\text{exc}} = \max(f_{ct} - b_{ct}, 0)$.

2.2 Event definition

An *outage event* begins when f_{ct}^{exc} exceeds 0.5% of county customers, must persist at least 2 hours, and merges with a neighbouring exceedance separated by less than 6 hours. Restoration is declared when the excess fraction decays to 10% of the event peak. Events still active at the end of the record are marked right-censored and treated as such by the duration model. For each county-day the event table yields the binary occurrence indicator Y_{ct} , the integrated consequence in customer-hours, the peak outage fraction, the restoration duration, and a concurrent statewide load term.

2.3 Atmospheric predictors

Hourly ERA5 reanalysis (Hersbach et al., 2020) supplies 10 m and 100 m winds, instantaneous 10 m wind gust, total precipitation, 2 m temperature, volumetric soil water in the upper two layers, convective available potential energy, snowfall, and snow depth over a Michigan domain. Forecast case studies use the NOAA Global Ensemble Forecast System version 12 (Zhou et al., 2022), 31 members at 0.25° , at lead times of 5, 3, 2, and 1 days.

2.4 Static exposure

County geometry comes from U.S. Census TIGER/Line. NLCD percent tree canopy cover (Coulston et al., 2012) provides a county-mean vegetation exposure proxy. We stress that canopy cover is an exposure proxy only: it carries no information about vegetation-management practice, trim cycles, or right-of-way condition, which are among the strongest real controls on storm outage rates and are not observable in any public dataset used here.

3 Methods

Figure 1 summarizes the pipeline. Each stage is described below.

3.1 Spatial aggregation and county-day features

The ERA5 0.25° grid is aggregated to county polygons in the CONUS Albers equal-area projection (EPSG:5070). Two aggregation operators are maintained rather than one, because the correct operator depends on the physics of the field. Smooth fields — temperature, soil moisture, accumulated precipitation — are aggregated with area weights that sum to unity per county. Localized damage agents — wind gust and CAPE — are aggregated with spatial maxima and upper percentiles over the cells intersecting each county. Using an area-weighted mean for gust is not a conservative choice but a destructive one: it averages the peak gust of a squall line together with calm cells at the far end of the county and removes precisely the signal that damages the network.

Hourly county series are then reduced to daily features: maximum and 95th-percentile gust, mean gust, maximum sustained and 100 m winds, total and maximum hourly precipitation, hours above gust thresholds of 13, 18, and 25 ms^{-1} , mean soil moisture, maximum CAPE, minimum and maximum temperature, a freezing-rain proxy (precipitation accumulated while 2 m temperature lies in $[-2, +0.5]^\circ\text{C}$), a wet-snow proxy (snowfall accumulated while temperature lies in $[-1, +1]^\circ\text{C}$), 7-day antecedent precipitation, and interaction terms coupling gust to canopy, soil moisture, and a leaf-on indicator. A heuristic regime label (convective wind, synoptic wind, ice, wet snow, benign) is attached for stratified verification; it is a diagnostic convenience, not ground truth.

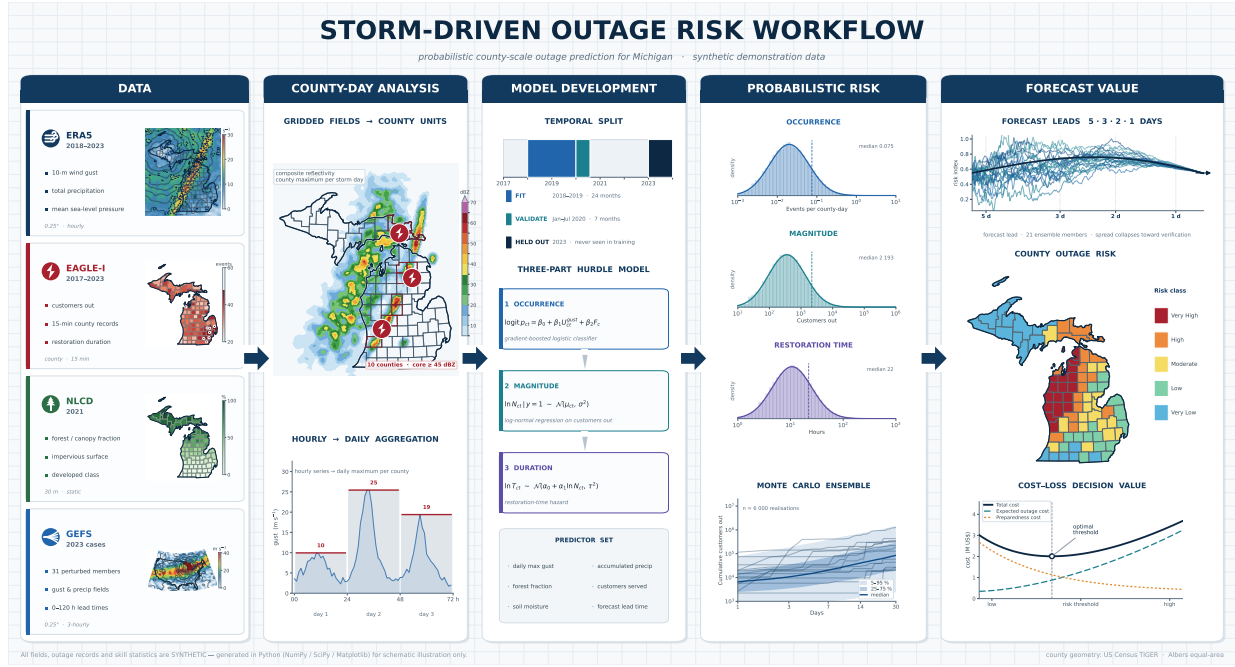


Figure 1. Analysis workflow. Hourly ERA5 reanalysis, EAGLE-I county outage records, NLCD canopy, and GEFS ensemble forecasts are aggregated to county-day features in an equal-area projection; a three-part probabilistic model produces county-day risk, and Monte Carlo composition supplies the predictive distributions used in the decision-value analysis. The maps, curves, and density shapes in this schematic illustrate the analysis design; empirical results appear in Section 4.

3.2 Three-part probabilistic model

For county c and day t the model estimates a predictive distribution of consequence, decomposed as a hurdle:

$$p_{ct} = P(Y_{ct} = 1 | \mathbf{X}_{ct}), \quad (1)$$

estimated by gradient-boosted decision trees (Friedman, 2001; Ke et al., 2017) under a binary objective with positive-class reweighting, then mapped to calibrated probabilities by isotonic regression (Zadrozny and Elkan, 2002) fitted on the validation window;

$$H_{ct} | Y_{ct} = 1 \sim F_{\mathbf{X}_{ct}}, \quad (2)$$

the conditional distribution of customer-hours, estimated in $\log(1 + \cdot)$ space by natural-gradient boosting of a Gaussian density (Duan et al., 2020), with independently fitted pinball-loss quantile models as a documented fallback; and

$$T_{ct} | Y_{ct} = 1 \sim \text{Weibull AFT}(\mathbf{X}_{ct}), \quad (3)$$

the conditional restoration duration, fitted as an accelerated failure time model that treats unresolved events as right-censored (Kalbfleisch and Prentice, 2002), with a Cox proportional-hazards fit retained as a diagnostic.

Reported quantiles are $\{0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95\}$. Where independently fitted quantile regressions cross, monotonicity is restored by rearrangement, and the raw crossing count is recorded as evidence about how hard the quantile route is straining.

3.3 Monte Carlo composition

The three stages are composed by Monte Carlo: for each county-day, 1000 draws sample occurrence from (1), and for the draws that fire, magnitude from (2) and duration from (3). Composition rather than multiplication of conditional means is what preserves the zero inflation and the heavy right tail; a composed median is not the median of the composition, and the difference is largest exactly where the decision matters.

3.4 Verification framework

Absolute scores are uninformative without reference forecasts, so every occurrence metric is reported against three: *county* climatology (the county’s own historical event rate), a reduced-feature logistic GLM, and a 20 m s^{-1} gust-threshold rule. County climatology, not a single statewide rate, is the correct reference: Michigan county event rates vary by an order of magnitude, and skill measured against a pooled rate flatters the model. Magnitude is referenced to the training climatological quantiles of customer-hours, and restoration to county-specific persistence (the county’s historical median restoration time).

Occurrence is scored with the Brier score (Brier, 1950) and its skill relative to county climatology, average precision, ROC AUC, and log loss. Magnitude is scored with the continuous ranked probability score (Gneiting and Raftery, 2007), computed as the pinball-loss integral over a 999-point probability grid with probit-slope tail extrapolation — not as a truncated ensemble approximation, which discards the outer predictive mass where most customer-hours live —

Table 1. Frozen temporal design. The split was fixed before the data were examined and was not revisited.

Period	Dates	Role
Training	2018-01-01 – 2022-06-30	Fit the model
Validation	2022-07-01 – 2022-12-31	Calibrate and verify
Test	2023-01-01 – 2023-12-31	One-time evaluation

together with a logarithmic score and a probability integral transform histogram (Gneiting et al., 2007). Restoration is scored with Harrell’s concordance index (Harrell et al., 1982) and median absolute error on uncensored events.

Reported validation probabilities are *out-of-fold* within the validation window. Isotonic regression is flexible, and scoring a calibrator on the rows it was fitted on makes every occurrence metric and every reliability diagram look better than the model is; the in-sample value is retained alongside so the optimism remains visible rather than being reported as the result.

3.5 Cross-validation and leakage control

Adjacent county-days within one storm are not independent observations. Random folds place the same storm on both sides of a split and reward memorising storms. We therefore block cross-validation by *storm episode*: a storm day is one on which at least 10% of reporting counties are in event, consecutive storm days are merged across gaps of up to two days, and each day is assigned to its nearest episode. Defining a storm day as one on which *any* county is in event does not work at this cohort size — with a per-county event rate of a few percent, almost every calendar day qualifies and the entire multi-year record fuses into a single group. Leave-one-county-out and forward-year folds are retained alongside storm blocking as complementary diagnostics of spatial and temporal generalization.

3.6 Experimental design

Table 1 gives the frozen split. All timestamps are UTC. The fitted model, the isotonic calibrator, the reporting cohort, and a SHA-256 hash of the full configuration are serialized together; the test evaluation refuses to run if the configuration hash has changed or if the rebuilt test-year cohort differs from the frozen one. A marker file records that the test year has been opened, and is written only after scoring has succeeded.

An earlier January–May 2021 retrospective is superseded and not reported here, because 2021 is now training data; forward-year cross-validation is the appropriate temporal diagnostic in the final design.

4 Results

4.1 Occurrence

On the 2023 holdout the occurrence model attains a Brier score of 0.0795 (Table 2), improving on all three references: county climatology (0.0859), the reduced logistic

GLM (0.0840), and the gust threshold rule (0.1118). The Brier skill score against county climatology is +0.075, down from +0.101 in validation. Discrimination is stable across the two windows — ROC AUC 0.746 on test against 0.753 on validation — while average precision falls from 0.339 to 0.277.

The pattern is worth stating plainly, because the absolute Brier score *improved* from validation to test while the skill score *declined*. Both are correct and neither is the headline on its own: 2023 was an easier year to forecast in absolute terms, and the climatological reference improved along with the model. Skill relative to climatology is the quantity that generalizes, and it remained positive but smaller. That the margin over a nine-feature logistic GLM is modest (0.0795 versus 0.0840) is itself a result: on this target, most of the achievable occurrence skill is captured by a small set of interpretable hazard and exposure predictors, and the gradient-boosted model adds a real but incremental improvement.

Figure 2 collects the reliability curve, precision–recall curve, magnitude PIT histogram, reference-model comparison, regime stratification, and cross-validation summary for the frozen model.

4.2 Magnitude and restoration

Magnitude skill relative to the climatological quantiles of customer-hours is essentially unchanged between windows (+0.083 validation, +0.084 test), but the absolute errors are not: CRPS rises from 16,011 to 39,164 and median RMSE from 146,422 to 501,575 customer-hours. The stability of the skill score alongside a threefold rise in absolute error indicates that 2023 contained larger events than the validation half-year rather than that the magnitude model degraded; the climatological reference deteriorated by a comparable factor. The very large median RMSE relative to CRPS is the expected signature of a heavy-tailed target: a proper score integrated over the full predictive distribution is far more forgiving of tail events than a squared error on a central estimate, which is one reason CRPS rather than RMSE is the headline magnitude metric here.

Restoration concordance falls from 0.663 to 0.619, and median absolute error rises from 4.42 to 5.22 hours. Notably, the county-persistence reference achieved 4.20 hours on the test year — *better* than the model’s 5.22. We report this without softening it: on the 2023 holdout, the restoration model did not beat a county-median persistence baseline on median absolute error, even though it ranked events better than chance (concordance 0.619). Restoration duration is driven by crew dispatch, mutual aid, and access constraints that no atmospheric predictor observes, and this result is the clearest quantitative evidence in the study for where utility operational data would add the most.

4.3 Spatial diagnostics

Figure 3 maps county-level event rate, mean forecast probability, bias, Brier skill, event count, and magnitude CRPS.

Table 2. Verification of the frozen model. The 2023 column is the primary result and was produced once, after freezing; validation is shown for context. Arrows give the direction of improvement. Reference rows are alternative forecasts, not model outputs.

Metric	Better	Validation	2023 test
<i>Occurrence</i>			
Brier score	↓	0.0926	0.0795
Brier skill vs. county climatology	↑	+0.101	+0.075
Average precision	↑	0.339	0.277
ROC AUC	↑	0.753	0.746
Log loss	↓	0.3196	0.2838
Ref.: logistic GLM (Brier)	↓	0.0999	0.0840
Ref.: county climatology (Brier)	↓	0.1030	0.0859
Ref.: gust > 20 ms ⁻¹ rule (Brier)	↓	0.1418	0.1118
<i>Magnitude (customer-hours given an event)</i>			
CRPS	↓	16,011	39,164
CRPS skill vs. climatology	↑	+0.083	+0.084
Logarithmic score	↓	9.242	9.003
Median RMSE	↓	146,422	501,575
<i>Restoration duration</i>			
Concordance index	↑	0.663	0.619
Median absolute error (h)	↓	4.42	5.22
Ref.: county persistence MAE (h)	↓	4.57	4.20

Frozen outage-risk model performance — held-out test · 2023

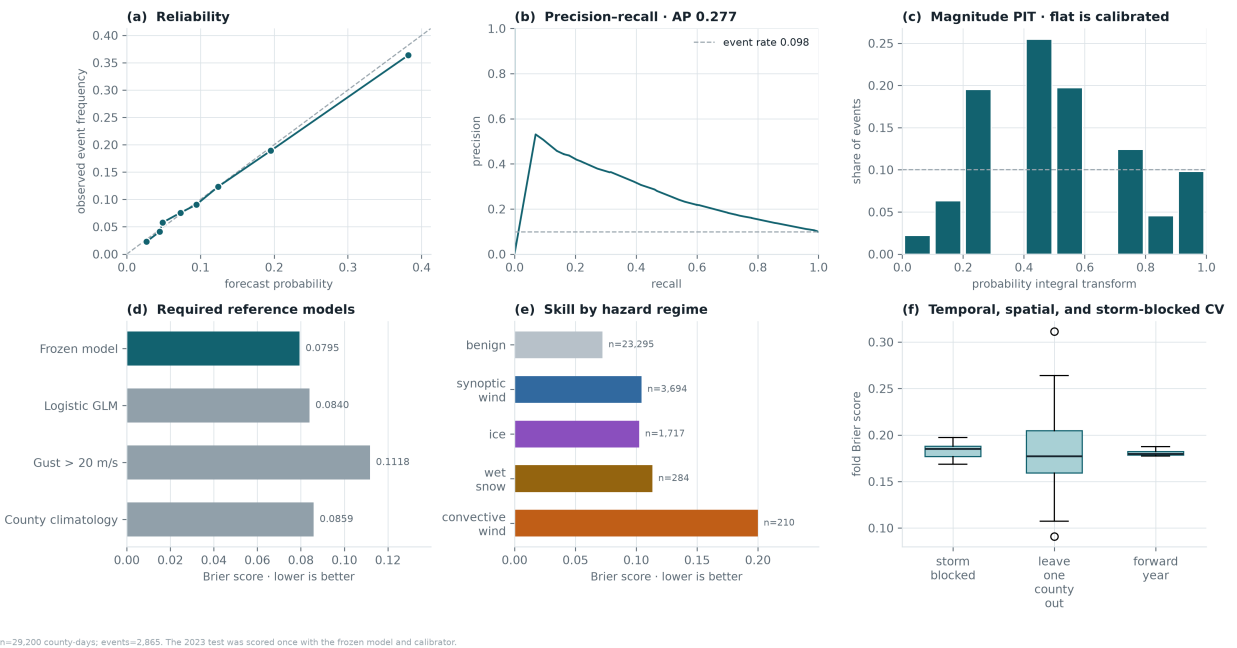
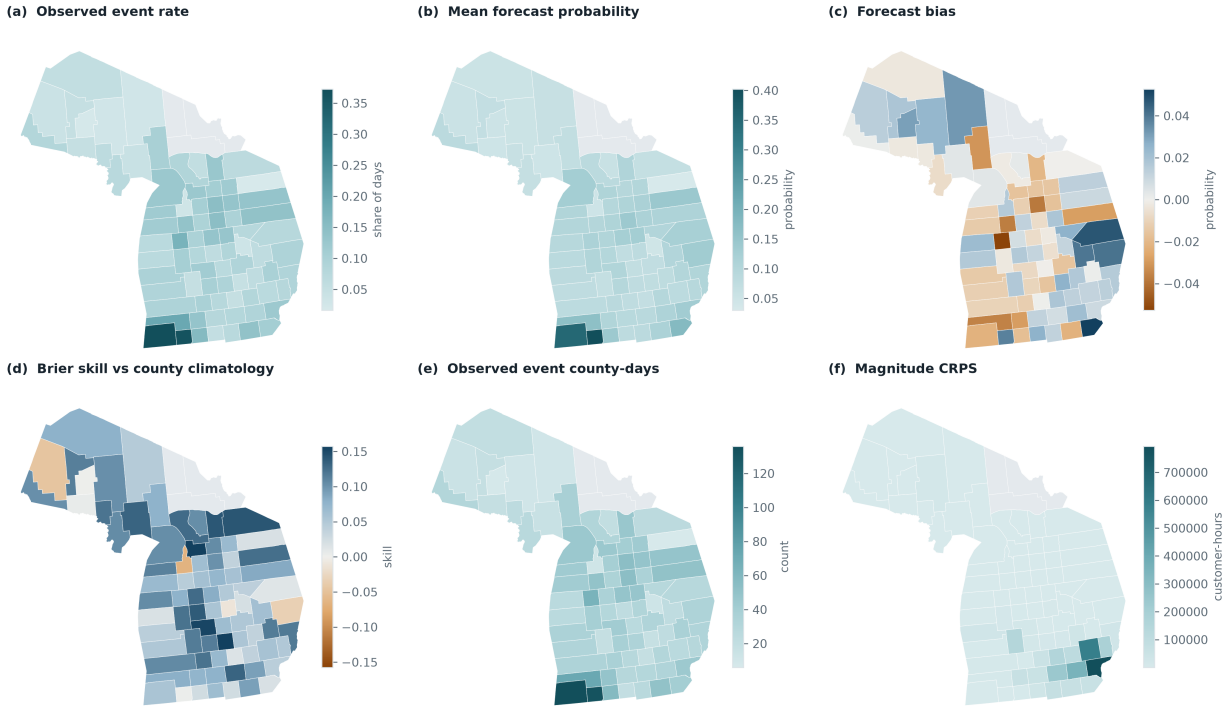


Figure 2. Frozen-model skill summary: occurrence reliability and precision–recall, magnitude probability integral transform histogram, comparison against reference forecasts, stratification by hazard regime, and cross-validation across storm-blocked, leave-one-county-out, and forward-year schemes. Reliability uses out-of-fold calibrated probabilities.

Skill is not spatially uniform, and the counties where it is weakest are informative: sparse-coverage and low-event-count counties provide few events to learn from, so their

county-specific climatology is itself poorly estimated and the skill score against it is correspondingly noisy.

County-level diagnostic results — held-out test · 2023



80 reporting counties. Orange/blue panels are centred on zero; grey counties have no evaluable data. County estimates with few events should be read as diagnostics, not rankings.

Figure 3. County-level diagnostics for the 2023 holdout: observed event rate, mean forecast probability, bias, Brier skill against county climatology, event count, and magnitude CRPS.

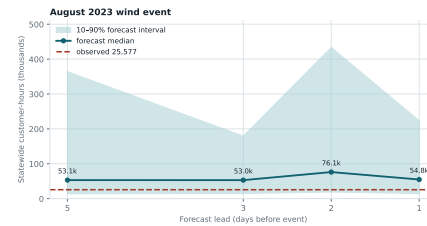
4.4 GEFS case studies

Each of the 31 GEFS members is bias-corrected against an ERA5 climatology built from training and validation months only, passed through the feature pipeline, and composed into 100 parametric draws, giving 3100 realizations per lead time. The quantile mapping is fitted once per case and applied unchanged across all leads; re-fitting it per lead forces every lead onto the same climatology and erases exactly the lead-dependent dispersion the case studies are designed to measure.

Table 3 and Figure 4 report the results. The two cases behave very differently. All four August wind-event intervals contained the observation of 25,577 customer-hours, with medians roughly a factor of two high and predictive intervals spanning an order of magnitude — wide, but honest. Every February ice-storm interval fell *below* the observed 38,402 customer-hours, at every lead. This is a genuine failure, not a wide interval: the framework did not merely underpredict the ice storm, it assigned the observed outcome negligible probability at all four lead times.

The variance decomposition adds a further caution. For the August case, the share of predictive variance attributable to ensemble spread rises with decreasing lead (9.6% at day −5 to 36.0% at day −1), the expected behaviour as the meteorological uncertainty becomes better resolved relative to

Probabilistic customer-hour forecast by GEFS lead



August wind case only. Lines are frozen-model medians; shading is the 10-90% predictive interval; dashed line is realized 2023 customer-hours. Forecast distributions use GEFS members × conditional draws.

Figure 4. Statewide customer-hours forecasts for the August 2023 wind event by GEFS lead time. Line: forecast median. Shading: 10–90% predictive interval. Dashed line: the 25,577 customer-hours observed.

the model’s fixed conditional uncertainty. For the February case the same share stays between 6.7% and 18.1% with no monotone trend, consistent with a hazard representation that is not responding to the ensemble information.

5 Decision value

Skill is a property of forecasts; value is a property of decisions. We evaluate value in two complementary ways: a dimensionless cost-loss analysis that requires no monetary assumption, and a scenario analysis that does.

Table 3. Statewide customer-hours forecasts driven by GEFS for the two 2023 case studies. Meteorological variance is the share of total predictive variance attributable to spread among ensemble members, as opposed to the model’s own conditional uncertainty. These are operational case diagnostics and do not replace the annual verification in Table 2.

Case	Lead	Median	10–90% interval	Observed	Met. variance
Aug 2023 wind event	day −5	53,113	13,677–364,016	25,577	9.6%
Aug 2023 wind event	day −3	52,981	14,710–179,218	25,577	11.1%
Aug 2023 wind event	day −2	76,129	18,787–433,563	25,577	26.8%
Aug 2023 wind event	day −1	54,824	14,710–223,482	25,577	36.0%
Feb 2023 ice storm	day −5	2,454	0–15,818	38,402	11.5%
Feb 2023 ice storm	day −3	4,272	0–20,890	38,402	18.1%
Feb 2023 ice storm	day −2	3,184	0–14,660	38,402	9.0%
Feb 2023 ice storm	day −1	3,462	0–16,146	38,402	6.7%

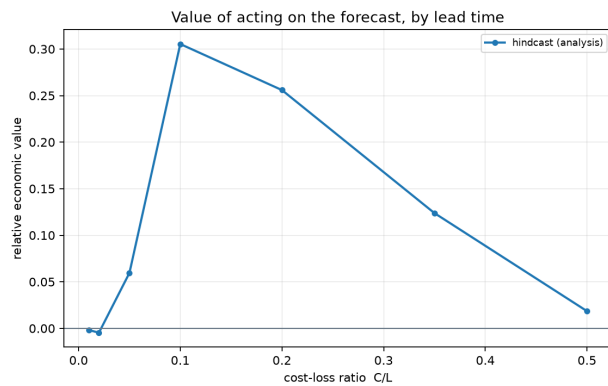


Figure 5. Relative economic value of the calibrated occurrence forecast across cost–loss ratios, against climatology and perfect information.

5.1 Cost–loss framework

In the standard cost–loss setting a decision maker can pay a protective cost C to avoid a loss L , and the optimal policy under a calibrated probability forecast is to act when $p \geq C/L$ (Richardson, 2000; Zhu et al., 2002). Relative economic value compares the expense of that policy against always acting on climatology and against perfect information, and is dimensionless: it depends on the ratio C/L , not on either quantity separately. Figure 5 reports relative value across the configured range $C/L \in [0.01, 0.50]$. This is the part of the value analysis that stands on its own, because it introduces no dollar figure at all.

5.2 Monetary scenarios and their standing

Note on monetary figures. The interruption-cost proxy below is a *documented placeholder*, constructed from a customer mix and per-hour values in the configuration file, in the spirit of the ICE Calculator framework (Sullivan et al., 2015). It is not measured Michigan damage, repair expense, or utility financial loss, and every avoided-cost figure is a counterfactual under an assumed mitigation effectiveness, not an observed saving. These numbers illustrate the arithmetic of the decision chain; they must be replaced with region- and utility-specific inputs before any external economic claim.

The configured proxy is \$25.12 per customer-hour, from a mix of 88% residential at \$4.00 and 12% commercial at \$180.00 per customer-hour. Applied to the 126,927,110 customer-hours of interruption observed across the 2023 test year, that proxy corresponds to approximately \$3.19 billion — a figure whose only purpose here is to establish the scale at which even small proportional reductions matter.

5.3 Forecast-triggered action scenarios

To make the decision chain concrete, consider the August wind event at a five-day lead. At each lead the model supplies a county probability p_i and a conditional customer-hours distribution. A county is flagged when $p_i \geq C/L = 0.10$; at day −5, 52 of the 80 counties in the case cohort crossed that threshold. Depending on lead time, the corresponding action package could include pre-positioning crews and materials, requesting mutual aid, inspecting vulnerable circuits, clearing vegetation, preparing switching plans, and customer communication.

After the event, the trigger mask is compared against what was observed. Total observed loss was 25,577 customer-hours; the triggered counties contained 25,285 customer-hours, or 99% of the event loss. Because no record of actual interventions exists, the scenario assumes an action reduces realized consequence by 20%:

$$\text{avoided customer-hours} = 25,285 \times 0.20 = 5,057, \quad (4)$$

$$\text{avoided proxy} = 5,057 \times \$25.12 \approx \$127,032. \quad (5)$$

Table 4. Forecast-triggered action for the August 2023 wind event at a cost–loss threshold of $C/L = 0.10$ and an assumed 20% consequence reduction. Values are counterfactual potential impacts, not observed savings. February ice-storm rows are omitted because no county crossed the trigger.

Forecast lead	Counties triggered	Observed loss covered	Potential avoided customer-hours	Potential avoided cost proxy
–5 days	52 / 80	99%	5,057	\$127,032
–3 days	45 / 80	95%	4,874	\$122,435
–2 days	50 / 80	99%	5,057	\$127,032
–1 day	40 / 80	73%	3,753	\$94,270

Table 4 repeats this across leads. Coverage is high and remarkably flat from day –5 to day –2 (99%, 95%, 99%) and falls at day –1 to 73% with fewer counties triggered. The non-monotonicity is itself worth noting: for this event, a five-day forecast supported an action footprint at least as useful as a one-day forecast, which is the operationally relevant finding, since five days is where mutual aid and crew staging decisions are actually made.

The counterfactual should not be over-read. The estimate says that had a 20%-effective action been taken in the triggered counties, roughly 5,057 customer-hours would have been avoided. It does not say that such an action exists, that it costs less than its benefit, or that 20% is the right effectiveness. Establishing net benefit requires utility-specific action costs and intervention outcome records, neither of which is public.

6 Discussion

6.1 What the results support

Three claims are supported by the frozen evaluation. First, weather-conditioned county-day outage occurrence over Michigan is predictable with positive skill against county climatology, and that skill survived a genuine one-shot hold-out (Brier skill +0.075 in 2023). Second, the conditional consequence distribution carries positive skill against a climatological distribution (CRPS skill +0.084), meaning the model conveys information about *how large* an event will be and not only whether one occurs. Third, the calibrated probabilities are usable in a cost–loss decision rule, and in the August case study the resulting trigger mask covered nearly all realized loss at operationally useful lead times.

6.2 The ice-storm failure

The February ice storm is the most informative negative result in the study, and it should not be filed as a wide interval. Every lead time placed the observed consequence outside its 10–90% interval, and the meteorological share of predictive variance showed no lead-dependent structure. Two explanations are consistent with the design and cannot be separated with the data at hand. The freezing-rain predictor is a proxy — precipitation accumulated within a temperature band — and does not resolve ice accretion, which depends on surface and near-surface thermodynamic structure a 0.25° field represents only crudely. Alternatively

or additionally, ice events are rare in the training record, so the magnitude model has few analogues from which to learn the appropriate conditional tail. Both point in the same direction: ice is the regime where this framework is least trustworthy, and reporting it as such is more useful than an aggregate score in which one rare regime is averaged away.

6.3 Limitations

Spatial support of the target. County aggregation hides feeder topology, asset age, conductor type, vegetation-management practice, and utility operations. A county is not a physical unit of the distribution network, and no result here should be read as an asset-level statement.

Consequence without mechanism. EAGLE-I records what customers experienced, not why. The model can say that outage consequence is likely; it cannot attribute that consequence to a cause.

Hazard resolution. ERA5 at 0.25° under-resolves convective gusts, and the freezing-rain and wet-snow terms are thermodynamic proxies rather than accretion models.

Case-study sample size. Two case studies cannot characterize forecast performance. The GEFS bias correction is fitted on a small sample, and the case results are diagnostics, not verification.

Restoration. The duration model did not beat county persistence on median absolute error in 2023. Restoration is an operations problem more than a meteorological one.

Economics. All monetary figures rest on placeholder interruption costs and an assumed mitigation effectiveness, and the asset-count denominator behind any per-asset break-even figure is a customer-count ratio, not a pole and span inventory.

6.4 Future work

The most valuable additions are not more model capacity but better data: utility-specific interruption costs and intervention records would convert the counterfactual scenarios into measured value; feeder-level outage records would allow the target to be defined on a physical unit of the network; and a higher-resolution or convection-permitting hazard representation would address the gust and ice-accretion limitations directly. On the modelling side, the regime-stratified results

suggest that a hazard-specific treatment of ice, with an explicit accretion formulation, would address the study’s clearest failure mode more effectively than a general increase in model flexibility.

7 Conclusions

We have presented a county-day probabilistic framework linking reanalysis and ensemble forecasts to distribution-system outage consequence over Michigan, and evaluated it under a frozen split scored once on a full held-out year. The occurrence model beats county climatology, a reduced logistic GLM, and a gust-threshold rule on the 2023 holdout; the conditional magnitude distribution carries positive skill against climatology; and the calibrated probabilities support a cost-loss decision rule whose trigger mask covered 99% of realized loss at a five-day lead in the August case study. Equally, the framework did not beat county persistence for restoration duration, and it failed on the February ice storm at every lead time. Reporting both outcomes from a single frozen evaluation is the point of the design: the framework is useful for wind-driven outage anticipation at operational lead times, and it is not yet reliable for ice.

Data availability

EAGLE-I outage records are distributed by Oak Ridge National Laboratory (Brelsford et al., 2024) via figshare (10.6084/m9.figshare.24237376). ERA5 is available from the Copernicus Climate Data Store and the ECMWF ARCO archive. GEFS is available from the NOAA public cloud archive. NLCD percent tree canopy cover is distributed by the Multi-Resolution Land Characteristics Consortium. County geometry is from U.S. Census TIGER/Line.

Code availability

The complete analysis pipeline, configuration, and reporting code are available at github.com/afahadabdullah/storm-outage-risk-mi. The study region, temporal splits, event definition, and hazard thresholds are set in a single configuration file, so the framework can be moved to another region without modifying analysis code.

Competing interests

The author declares no competing interests.

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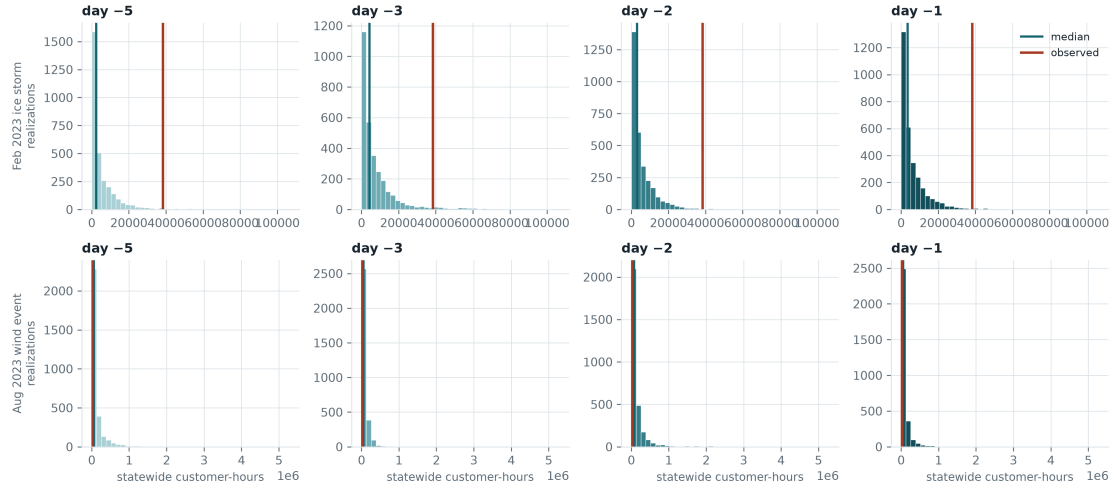
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A Supplementary figures

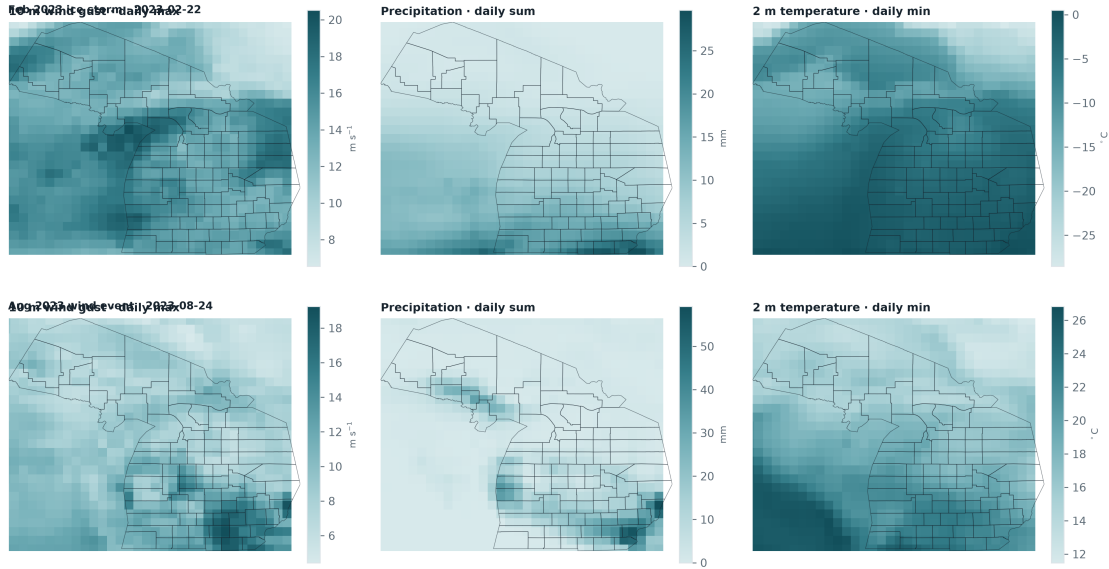
Frozen-model GEFS case-study forecasts



31 GEFS members $\times$ 100 conditional model draws per lead. Gust and precipitation are quantile-mapped to the pre-test ERA5 climatology. Observed lines appear only after the once-only 2023 test is opened.

Figure 6. GEFS-driven predictive distributions of statewide customer-hours for both 2023 case studies at all four lead times. The August wind event (contained at every lead) and the February ice storm (below the observation at every lead) are shown on comparable axes.

ERA5 hazard fields on frozen-model GEFS case-study days



Native ERA5 grid cells; gust/temperature are daily extrema and precipitation is daily accumulation. County outlines are shown for spatial context.

Figure 7. ERA5 hazard fields on the two case-study days, showing the meteorological setting the model conditioned on in each case.

August 2023 wind event · forecast-triggered action by lead

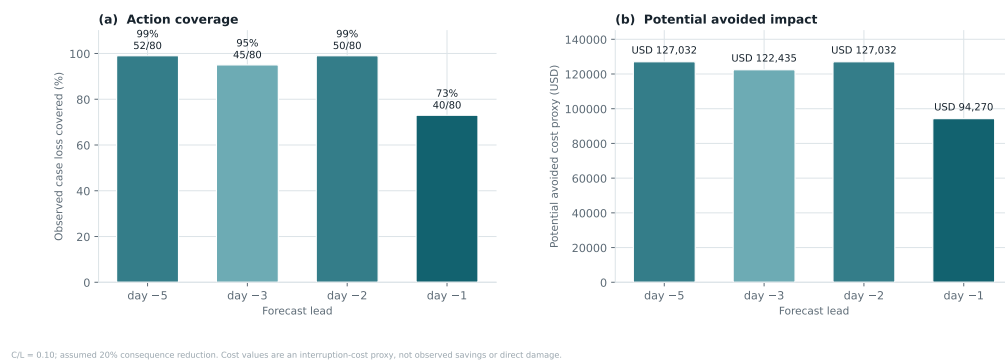


Figure 8. Forecast-triggered action for the August 2023 wind event by lead time: share of observed loss contained within triggered counties, and the potential avoided interruption-cost proxy under the 20% mitigation assumption. Labels above the coverage bars give triggered counties out of 80. All monetary values are placeholders as described in Section 5.